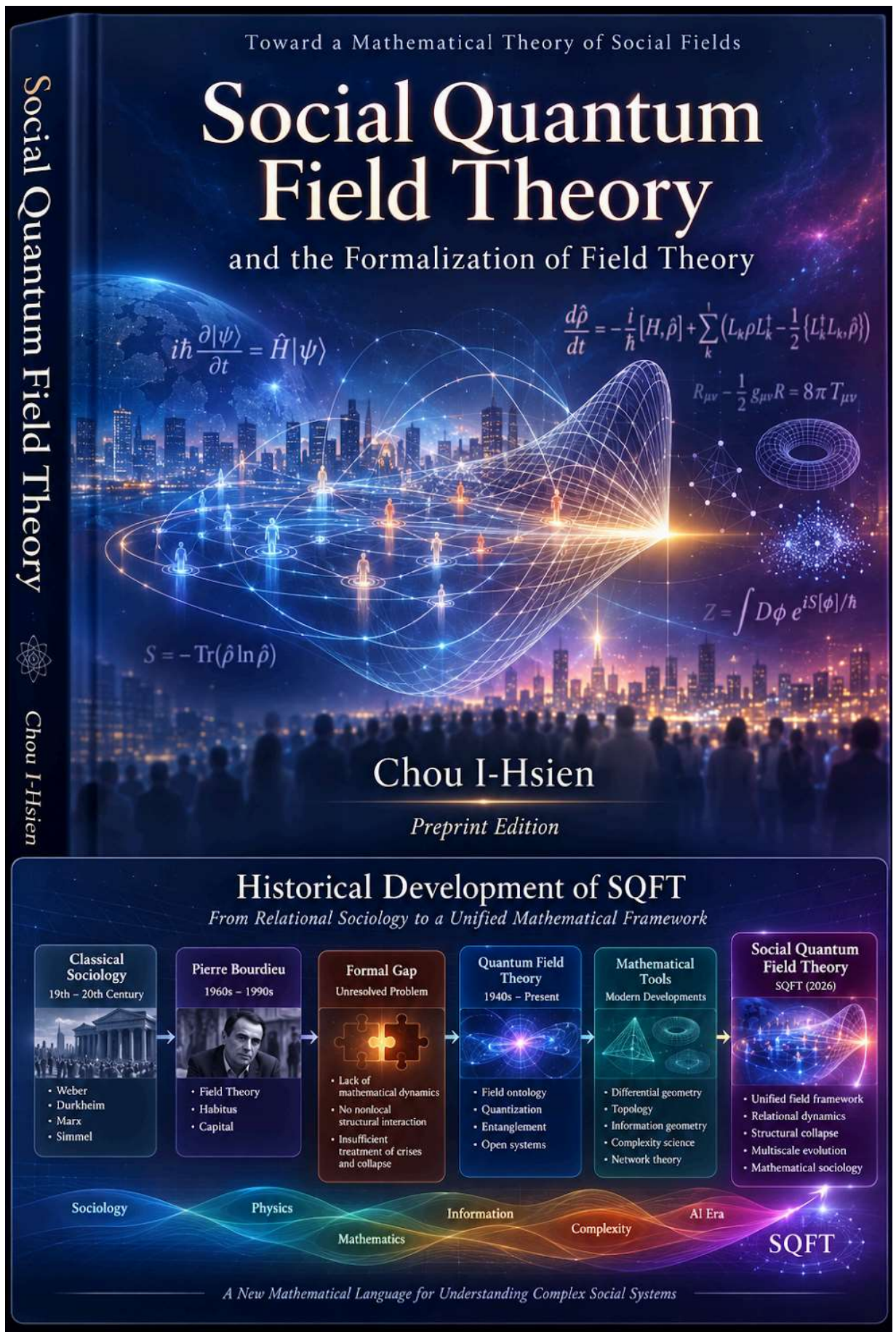


**Social Quantum Field Theory. First Edition (2026) 1~10
Volume I**



7月 19, 2026



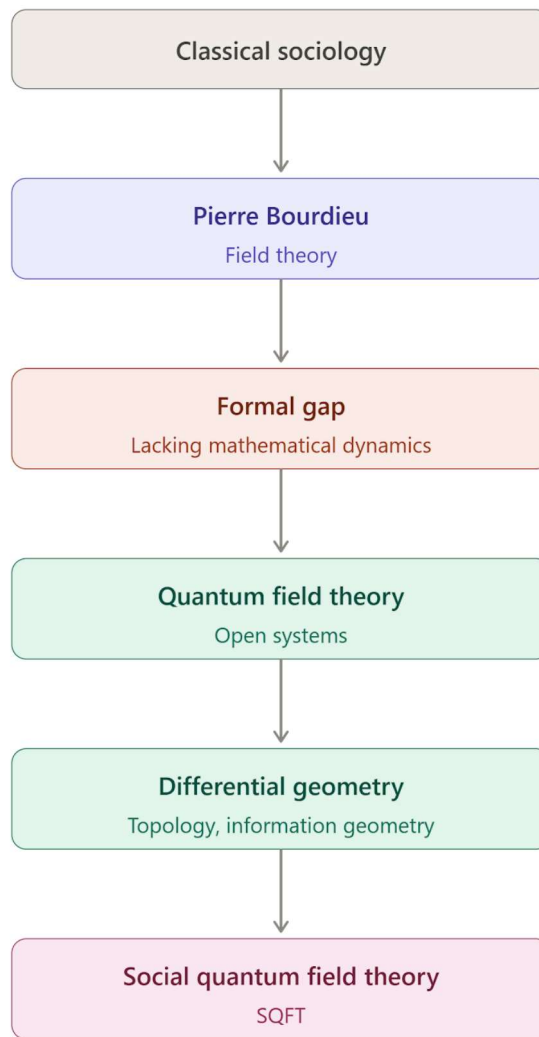


Figure 1.1 Conceptual lineage from classical sociology and Bourdieu's field theory to the formal mathematical framework of Social Quantum Field Theory.

Social Quantum Field Theory

Foundations of Social Quantum Field Theory

Chapter 1

Introduction

1.1 The Motivation

Modern science has achieved remarkable success in describing the physical universe through mathematically precise theories. Classical mechanics explains planetary motion, quantum mechanics describes microscopic systems, and quantum field theory unifies particles and interactions within a common mathematical language.

By contrast, the mathematical description of complex social systems remains comparatively fragmented. Economics, sociology, political science, and network theory each provide valuable perspectives, yet no widely accepted mathematical framework exists for representing **relational structures** as fundamental dynamical objects.

This monograph proposes **Social Quantum Field Theory (SQFT)** as one possible step toward such a framework.

The objective is not to import the laws of physics into sociology.

Instead, SQFT borrows mathematical structures that have proven useful for describing highly interconnected systems and adapts them into a formal language for relational dynamics.

1.2 The Central Question

The central question addressed throughout this volume is

Can relational structures themselves be regarded as primary mathematical objects, from which observable social behavior emerges?

Traditional mathematical models generally begin with independent agents whose interactions generate collective behavior.

SQFT reverses this order.

Instead of beginning with isolated actors,

it begins with a relational field.

Individuals become localized manifestations of that field.

1.3 Historical Background

Relational thinking has appeared repeatedly throughout intellectual history.

Examples include

- differential geometry describing space through local structure,
- gauge theory describing interactions through symmetry,

- network science emphasizing connectivity,
- category theory emphasizing morphisms rather than objects,
- Pierre Bourdieu's field theory emphasizing relational positions rather than isolated individuals.

SQFT draws inspiration from these traditions while remaining mathematically independent of any particular discipline.

1.4 Scope of the Framework

Throughout this monograph,

the word **field** does **not** necessarily denote a physical quantum field.

Instead,

a field refers to a mathematical object representing the relational organization of a complex system.

Possible applications include

- social institutions,
- financial markets,
- scientific communities,
- technological ecosystems,
- organizational networks,
- cultural systems.

The same mathematical framework may admit different interpretations depending upon the application.

1.5 Methodological Position

This work adopts four methodological principles.

Principle 1

Mathematics precedes interpretation.

Definitions and theorems should remain valid independently of any particular application.

Principle 2

Physical terminology is employed only when mathematically useful.

No claim is made that human society literally obeys microscopic quantum mechanics.

Principle 3

Every mathematical construction should remain open to comparison with existing methods.

Principle 4

Computational implementation should remain reproducible whenever possible.

1.6 Structure of the Book

The book develops progressively.

Part I

Foundational concepts

Part II

Axiomatic construction

Part III

Operator dynamics

Part IV

Open systems

Part V

Geometry and topology

Part VI

Computational framework

Part VII

Open mathematical problems

1.7 Contribution

The principal contributions of this work are

1. a relational mathematical ontology;

2. an axiomatic foundation;
3. a field-based variational formulation;
4. an operator-theoretic dynamical framework;
5. an information-geometric interpretation;
6. computational methods suitable for future numerical implementation.

Whether these constructions ultimately prove useful remains an open question for future mathematical investigation.

1.8 Reading Guide

Readers from different backgrounds may prefer different entry points.

- Mathematicians may begin with Chapters 3–8.
- Physicists may focus on operator methods and variational principles.
- Computer scientists may emphasize computational chapters.
- Sociologists may first read Chapters 1–2 before proceeding to the formal development.

Although inspired by multiple disciplines, the framework should ultimately be evaluated according to mathematical consistency rather than disciplinary origin.

Chapter Summary

This chapter introduced the motivation, scope, methodology, and objectives of Social Quantum Field Theory.

The following chapter reviews existing relational approaches and identifies the mathematical gap that motivates the SQFT framework.

Editorial Note

This revised version adopts the conventions of an international mathematical monograph. Throughout the remainder of the book, each chapter concludes with a summary, and all formal mathematical statements (Definitions, Axioms, Theorems, Propositions, Corollaries, and Remarks) are presented in a standardized style for consistency and ease of reference.

Chapter 2

From Bourdieu's Field Theory to Social Quantum Field Theory

2.1 Introduction

Few sociological theories have reshaped the understanding of social structure as profoundly as Pierre Bourdieu's theory of practice.

Rejecting both methodological individualism and rigid structural determinism, Bourdieu proposed that social reality is fundamentally **relational**. Individuals do not exist as isolated decision-makers; rather, they occupy positions within historically constituted fields whose internal relations define opportunities, constraints, and symbolic value.

This insight represents one of the most significant conceptual advances in twentieth-century sociology.

Nevertheless, an important mathematical question remains unresolved:

How can a relational field simultaneously organize the behavior of many actors without reducing that organization to chains of pairwise interactions?

The present chapter argues that this unresolved question motivates the development of Social Quantum Field Theory (SQFT).

2.2 The Historical Achievement of Bourdieu

Bourdieu introduced three foundational concepts:

- **Field**
- **Capital**
- **Habitus**

Together they replaced substance-based sociology with relational sociology.

Rather than asking

Who possesses power?

Bourdieu instead asked

What relational position makes power meaningful?

Thus,

power,

prestige,

authority,

knowledge,

and legitimacy

are understood as properties emerging within a relational structure.

This shift transformed sociology from the study of isolated actors into the study of structured relational spaces.

2.3 Fields as Structured Spaces

A field is not merely an environment.

Instead,

it is a structured configuration of positions.

Mathematically we may write

$$F = (V, E, W),$$

where

- V denotes actors,
- E denotes relational links,
- W denotes weighted institutional relations.

This graph representation is useful,

but remains incomplete.

Graphs describe connectivity.

They do not describe continuous field dynamics.

2.4 Capital as a Relational Quantity

Bourdieu emphasized that capital possesses value only through recognition.

Economic capital,

symbolic capital,

cultural capital,

and social capital

derive their significance from the surrounding field.

Hence,

SQFT interprets capital as

$$C = C(F).$$

Capital is therefore

not an intrinsic property,

but a field-dependent variable.

A change in field structure may alter capital without altering the individual.

Example 2.1

A prestigious university degree possesses different social value in

- academia,
- finance,
- politics,
- entrepreneurship.

The diploma remains identical.

The surrounding field changes.

Therefore,

$$\Delta C \neq 0,$$

2.5 Habitus and Internalization

Habitus explains how objective structures become embodied dispositions.

However,

once habitus has formed,

a mathematical difficulty appears.

Suppose two actors possess similar habitus.

How do they remain synchronized over time?

Traditional explanations invoke

- imitation,
- communication,

- repeated interaction,
- institutional rules.

These mechanisms are important,
yet they generally operate **after** interaction occurs.

2.6 The Remaining Relational Gap

Consider experienced football players.

They frequently execute coordinated passes without verbal communication.

Their coordination appears immediate.

The conventional sequential model is

$$A \rightarrow B \rightarrow C.$$

SQFT proposes an alternative representation.

Instead,

both players respond to the same relational field,

$$F \rightarrow (A, B).$$

The pass does not create coordination.

It reveals coordination already encoded within the field.

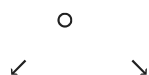
Figure 2.1

Classical

Player A -----> Player B

SQFT

Social Field



Player A Player B

Figure 2.1.

Sequential interaction versus field-mediated coordination.

2.7 Interaction versus Configuration

This distinction is fundamental.

Interaction asks

"How does one actor influence another?"

Configuration asks

"How does one field simultaneously constrain every actor?"

These are different mathematical problems.

The latter naturally motivates continuous field descriptions.

2.8 From Graphs to Fields

Graphs describe

$$(V, E).$$

Fields introduce

$$\Phi(x, t).$$

The field evolves continuously,

whereas graphs usually describe discrete topology.

SQFT therefore replaces

$$\text{Graph} \rightarrow \text{Field}.$$

Graph $\rightarrow$ Field.

Graphs become one possible discretization of an underlying relational manifold.

2.9 Field Ontology

The conceptual transition may be summarized as

Classical Sociology **SQFT**

Individual $\rightarrow$ Relation Relation $\rightarrow$ Individual

Classical Sociology	SQFT
Attributes	Field Configuration
Interaction	Continuous Dynamics
Network	Relational Manifold
Position	Local Excitation

This inversion is the defining ontological shift of SQFT.

2.10 Central Proposition

Proposition 2.1 (Field Primacy).

Observable social behavior is more naturally represented as the evolution of a relational field than as the aggregation of independent actors.

This proposition serves as a methodological principle rather than an empirically established theorem.

Its value depends upon whether subsequent mathematical constructions provide greater explanatory or computational power.

2.11 Transition to SQFT

The previous discussion suggests that relational sociology requires

- continuous dynamics,
- geometric structure,
- operator evolution,
- information flow,
- topology,
- symmetry.

These mathematical ingredients are largely absent from classical field theory in sociology.

The next chapter therefore introduces the axiomatic foundations of Social Quantum Field Theory.

Chapter Summary

This chapter reviewed Bourdieu's principal contributions while identifying the remaining formal gap: the absence of a mathematical description capable of representing simultaneous, field-mediated coordination among multiple actors.

By reinterpreting social relations as manifestations of a continuous relational field, SQFT shifts the analytical focus from interactions between individuals to the evolution of the field

itself. This transition establishes the conceptual bridge between relational sociology and the mathematical formalism developed in the following chapters.

Editorial Note

The constructions introduced in this chapter are intentionally **conceptual** rather than **empirical**. They motivate the mathematical framework but do not, by themselves, establish its validity. Subsequent chapters progressively replace qualitative arguments with formal definitions, axioms, propositions, and operator-based formulations that constitute the mathematical core of Social Quantum Field Theory.

Chapter 3

The Mathematical Philosophy of Social Quantum Field Theory

3.1 Why Another Mathematical Language?

The history of mathematics repeatedly demonstrates that scientific progress often begins with the development of a new mathematical language rather than the discovery of new empirical facts. Differential calculus transformed mechanics, tensor calculus enabled general relativity, and Hilbert spaces provided the natural setting for quantum mechanics. In each case, the mathematical formalism revealed structures that had remained inaccessible within earlier frameworks.

Social systems present a comparable challenge. Modern sociology possesses sophisticated qualitative concepts—such as field, capital, habitus, institution, and norm—but lacks a unified mathematical language capable of representing their continuous evolution, mutual dependence, and structural transformation.

The purpose of Social Quantum Field Theory (SQFT) is therefore not to replace existing sociological theories. Instead, it seeks to provide a formal mathematical framework in which relational structures can be defined, manipulated, and compared with the same degree of precision that geometry provides for space or operator theory provides for dynamical systems.

3.2 Mathematical Models and Physical Reality

A mathematical model is not identical to the system it represents.

Throughout the history of science, identical mathematical structures have frequently been applied to entirely different physical domains. Fourier analysis describes sound waves, heat conduction, electromagnetic fields, and financial time series. Differential geometry applies equally to planetary motion, robotic navigation, and machine learning.

Accordingly, SQFT adopts selected mathematical tools from quantum theory **without asserting that human society obeys microscopic quantum mechanics**.

This distinction is fundamental.

Definition 3.1 (Formal Analogy).

A *formal analogy* is a correspondence between mathematical structures that preserves relationships among variables while remaining agnostic about the underlying physical ontology.

Thus, the use of Hilbert spaces, operators, density matrices, or Lindblad equations within SQFT should be interpreted as the adoption of a mathematical language rather than a statement about the microscopic constitution of society.

3.3 From Objects to Relations

Classical scientific models often begin with objects.

Relations are introduced only after the objects have been defined.

Formally,

Object $\rightarrow$ Relation.

Object $\rightarrow$ Relation.

SQFT reverses this order.

The primary mathematical entity is the relational field itself.

Individual actors emerge as localized manifestations of that field,

Relation $\rightarrow$ Object.

Relation $\rightarrow$ Object.

This inversion represents the central ontological shift of SQFT.

3.4 Three Levels of Description

To avoid conceptual confusion, SQFT distinguishes three complementary levels of description.

Level I – Empirical Level

Observable quantities such as

- institutions,
- organizations,
- elections,
- financial markets,
- football teams,

constitute the empirical domain.

Level II – Mathematical Level

The empirical system is represented by

- field variables,
- operators,
- state vectors,
- density operators,
- manifolds,
- topological invariants.

These objects possess mathematical meaning independently of any specific interpretation.

Level III – Interpretive Level

Only after the mathematical structure has been established do sociological interpretations become appropriate.

Different applications may therefore employ the same mathematical framework while assigning different empirical meanings to its variables.

3.5 Domain Restriction

SQFT does **not** claim that

- electrons exist within societies,
- human beings possess quantum wavefunctions,
- microscopic quantum superposition governs human decision-making.

Instead, it proposes that certain mathematical structures developed in modern physics provide useful abstractions for representing complex relational systems.

This restriction prevents category errors between mathematical formalism and physical ontology.

3.6 Central Proposition

Proposition 3.1 (Mathematical Independence).

The validity of an SQFT model depends solely upon its internal mathematical consistency, logical coherence, and empirical usefulness—not upon whether its formal language originated in theoretical physics, sociology, or any other discipline.

Chapter Summary

This chapter established the philosophical foundations of Social Quantum Field Theory. It clarified the distinction between mathematical analogy and physical interpretation, introduced the transition from object-centered to relation-centered ontology, and identified the relational field as the primary mathematical object of the theory.

These principles provide the conceptual basis for the formal axiomatic development presented in the following chapter.

Chapter 4

The Eight Axioms of Social Quantum Field Theory

4.1 Why an Axiomatic Foundation?

Every mature mathematical theory begins not with applications, but with axioms.

Euclidean geometry begins with primitive notions of points and lines.

Classical mechanics begins with Newton's laws.

General relativity begins with the geometric structure of spacetime.

Quantum mechanics begins with Hilbert spaces and linear operators.

Similarly, Social Quantum Field Theory (SQFT) requires a small set of foundational assumptions from which subsequent definitions, propositions, and computational models may be derived.

These axioms are **not empirical laws**.

Rather, they specify the mathematical ontology of the framework.

Their value depends not upon philosophical appeal, but upon their logical consistency, mathematical productivity, and empirical applicability.

4.2 Axiom I — Field Primacy

Statement

The relational field is the primary mathematical object of the theory.

Individual actors are represented as localized manifestations of that field.

Formally,

$$F \succ A_i,$$

where

- F denotes the relational field,
 - A_i denotes the i -th actor,
 - $\succ$ denotes conceptual priority rather than temporal causation.
-

Interpretation

Classical social models assume

$$A_i \rightarrow F.$$

SQFT reverses this relation,

$$F \rightarrow A_i.$$

The field defines the space of possible identities before particular actors occupy those positions.

Example

A "judge" exists only because a legal field already exists.

Without the institutional field,

the biological individual remains,

but the social role disappears.

Remark

This axiom concerns mathematical representation,

not metaphysical determinism.

4.3 Axiom II – Local Excitation

Statement

A social actor is interpreted as a localized excitation of the underlying relational field.

Symbolically,

$$A_i = \hat{\Phi}(x_i)|0\rangle,$$

where

- $\hat{\Phi}(x)$ denotes a field operator,
 - x_i denotes a relational coordinate,
 - $|0\rangle$ denotes the background relational state.
-

Interpretation

The identity of an actor depends upon

- institutional context,
- cultural norms,
- historical conditions,
- surrounding relations.

No actor possesses complete meaning independently of the field.

Example

The same individual may simultaneously appear as

- professor,
- parent,
- investor,
- citizen.

These are distinct local excitations within different relational fields.

4.4 Axiom III – Relational Inseparability

Statement

Relations are mathematically prior to isolated attributes.

For sufficiently coupled actors,

$$|\Psi_{AB}\rangle \neq |\psi_A\rangle \otimes |\psi_B\rangle.$$

Interpretation

The state of actor A cannot always be specified independently of actor B.

Meaning is relational.

Example

Currency possesses value only within a monetary system.

Remove the surrounding field,

and its symbolic meaning vanishes.

Remark

This expression represents structural inseparability,

not microscopic quantum entanglement.

4.5 Axiom IV – Emergent Capital

Statement

Capital is a field-dependent observable.

$$C = C(F).$$

Interpretation

Capital is not an intrinsic property.

Its value emerges through collective recognition.

Example

Academic reputation,

political legitimacy,

or technological expertise

may rapidly change after institutional transformation,

even though the individual remains unchanged.

Social Field Operator $\Phi^S(x)$

OperatorInterpretation

$\hat{C}$	Capital
$\hat{P}$	Power
$\hat{T}$	Trust
$\hat{I}$	Information

4.6 Axiom V – Structural Collapse

Statement

Large-scale institutional transitions are represented as discontinuous projections of the relational state.

$$\rho \longrightarrow P\rho P.$$

Interpretation

Elections,
financial crises,
scientific revolutions,
or regime changes
may reorganize the relational field.

Example

Following a financial crisis,
previously valuable forms of capital may become obsolete,
while new forms emerge.

Remark

This mathematical projection should not be interpreted as physical wave-function collapse.

4.7 Axiom VI – Topological Evolution

Statement

Field evolution may alter topology without requiring continuous metric deformation.

Symbolically,

$$T(F_t) \neq T(F_{t+\Delta t}),$$

where

$$T$$

denotes a topological invariant.

Interpretation

Institutional restructuring,

organizational fragmentation,

or technological disruption

may correspond to changes in field topology rather than smooth parameter variation.

Example

The emergence of the Internet fundamentally reorganized communication topology rather than merely increasing communication speed.

4.8 Axiom VII – Open-System Dynamics

Statement

No empirical social field is perfectly isolated.

Its evolution satisfies

$$\frac{d\rho}{dt} = -\frac{i}{\hbar}[H, \rho] + \sum_k \left(L_k \rho L_k^\dagger - \frac{1}{2} \{L_k^\dagger L_k, \rho\} \right).$$

Interpretation

External influences include

- media,
- technology,
- legislation,
- education,
- international events,
- demographic change.

These influences continuously modify the relational field.

Remark

The Lindblad equation is adopted as a **general mathematical structure** for irreversible statistical dynamics.

Its use does **not** imply microscopic quantum evolution.

4.9 Axiom VIII – Formal Analogy and Domain Restriction

Statement

Every mathematical construction introduced within SQFT is interpreted as a formal analogy unless independently established as an empirical theorem.

Interpretation

The framework distinguishes three domains:

1. mathematical structure,
2. computational implementation,
3. empirical interpretation.

These domains should never be conflated.

Example

A density matrix represents incomplete relational information.

It does **not** imply that citizens literally occupy quantum superpositions.

4.10 Compact Mathematical Statement

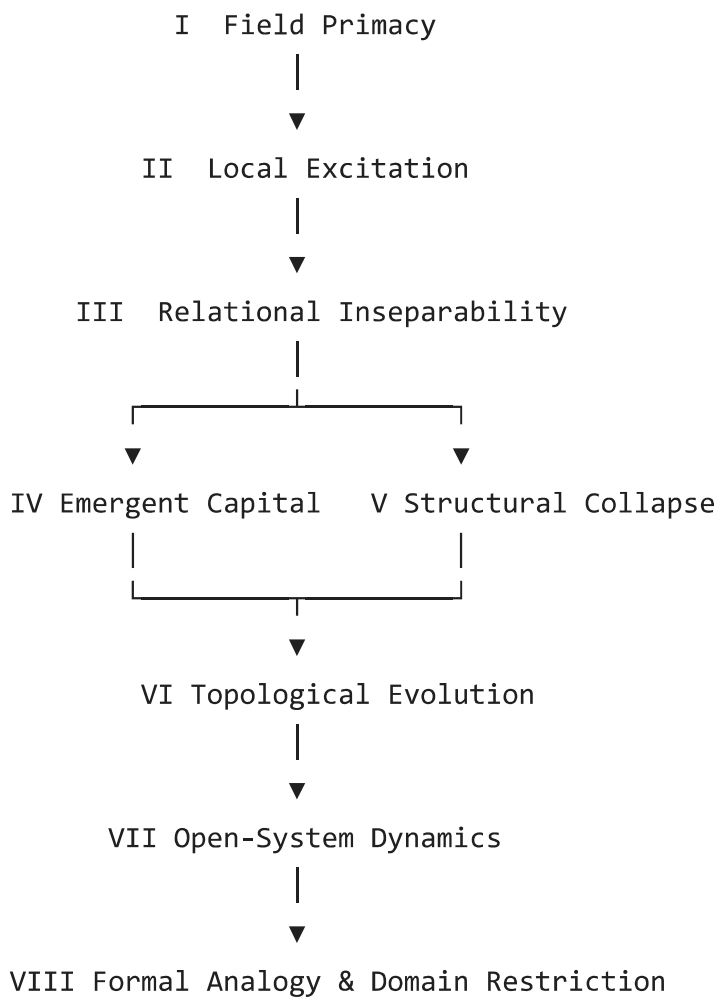
Collectively, the eight axioms may be summarized as

- I. $F \succ A_i$,
- II. $A_i = \hat{\Phi}(x_i)|0\rangle$,
- III. $|\Psi_{AB}\rangle \neq |\psi_A\rangle \otimes |\psi_B\rangle$,
- IV. $C = C(F)$,
- V. $\rho \rightarrow P\rho P$,
- VI. $T(F_t) \neq T(F_{t+\Delta t})$,
- VII. $\dot{\rho} = L(\rho)$,
- VIII. Formal Analogy $\neq$ Physical Identity.

I. $\mathcal{F} \succ A_i$	V. $\rho \rightarrow P\rho P$
II. $A_i = \hat{\Phi}(x_i) 0\rangle$	VI. $T(\mathcal{F}_t) \neq T(\mathcal{F}_{t+\Delta t})$
III. $ \Psi_{AB}\rangle \neq \psi_A\rangle \otimes \psi_B\rangle$	VII. $\dot{\rho} = \mathcal{L}(\rho)$
IV. $C = C(\mathcal{F})$	VIII. Formal Analogy $\neq$ Physical Identity

4.11 Internal Consistency

The eight axioms satisfy a logical dependency hierarchy:



This hierarchy is not intended as a proof but as a roadmap for the logical organization of the SQFT framework. The earlier axioms establish the ontology, the middle axioms describe field dynamics, and the final axiom specifies the methodological limits of interpretation.

Chapter Summary

This chapter established the **axiomatic foundation** of Social Quantum Field Theory. Beginning with the primacy of relational fields and ending with the restriction that all quantum-inspired constructions remain formal analogies unless independently validated, the eight axioms provide the conceptual and mathematical basis for the remainder of the monograph.

Subsequent chapters will no longer treat these principles as philosophical proposals. Instead, they will be used as assumptions from which definitions, operators, conservation laws, geometrical structures, and computational methods are systematically derived.

Editorial Note

In mature mathematical theories, axioms are judged not by their intuitive appeal but by the richness, consistency, and explanatory power of the structures they generate. The remainder of this book should therefore be understood as an exploration of the mathematical consequences of these eight axioms, rather than as independent philosophical arguments.

Chapter 5

The Quantum Entanglement Pass: A Worked Model of Relational Coordination

5.1 Introduction

The preceding chapters established the philosophical and axiomatic foundations of Social Quantum Field Theory (SQFT). A mathematical framework, however, acquires meaning only when its abstract concepts can be translated into concrete models.

This chapter presents the first complete worked example of SQFT.

The example is intentionally simple: a fictional football match in which two players appear to anticipate each other's actions without explicit communication. The scenario is inspired by the popular image of a "**Quantum Entanglement Pass**."

The objective is **not** to claim that football players become quantum-entangled in the physical sense.

Instead, the example illustrates how field-mediated relational coordination may be represented mathematically.

5.2 The Classical Passing Model

Traditional tactical analysis treats passing as a sequential process,

$$A \rightarrow B \rightarrow C.$$

Player A observes.

Player A decides.

Player A passes.

Player B reacts.

Coordination therefore emerges through successive information transfer.

This model works well for many situations.

However, elite teams often exhibit coordination that appears almost instantaneous.

Figure 5.1

Player A

|

▼

Pass

|

▼

Player B

Sequential information flow.

5.3 The Relational Field Model

SQFT proposes a different mathematical picture.

Rather than assuming

$$A \rightarrow B,$$

both players respond to a common relational field,

$$F \longrightarrow (A, B).$$

The pass does not generate coordination.

It reveals coordination already encoded within the field configuration.

Figure 5.2

Social Field

○



5.4 Joint Relational State

Instead of assigning independent states,

SQFT represents the team through a joint state,

$$|\Psi\rangle \in H_A \otimes H_B.$$

If

$$|\Psi\rangle = |\psi_A\rangle \otimes |\psi_B\rangle,$$

the players behave independently.

Otherwise,

$$|\Psi\rangle \neq |\psi_A\rangle \otimes |\psi_B\rangle.$$

The team's behavior cannot be reconstructed from either player alone.

Definition 5.1 (Relational Coordination)

A team exhibits **relational coordination** whenever its collective state is non-factorizable with respect to the participating actors.

5.5 Density Operator Representation

Real teams never possess perfect information.

Fatigue,

weather,

crowd pressure,

and incomplete tactical knowledge all introduce uncertainty.

Therefore,

the effective team state is represented by

$$\rho = \sum_i p_i |\Psi_i\rangle \langle \Psi_i|.$$

The density operator describes the statistical organization of the relational field.

5.6 Mutual Information

Coordination is quantified through mutual information,

$$I(A:B) = S(A) + S(B) - S(AB),$$

where

$$S(\rho) = -\text{Tr}(\rho \log \rho)$$

is the von Neumann entropy.

Large

$$I(A:B)$$

indicates strong relational dependence.

No direct communication is required.

Example 5.1

Two experienced midfield players frequently anticipate one another's movements.

Although no verbal instruction occurs,

their mutual information remains high because both continuously respond to the same tactical field.

5.7 Coach as an External Control Field

Classical tactics often describe coaching as issuing commands.

SQFT instead models the coach as an external control Hamiltonian.

The effective field becomes

$$H(t) = H_0 + H_{\text{coach}}(t),$$

where

- H_0 represents the intrinsic team dynamics,
- H_{coach} represents tactical intervention.

The coach does not directly determine every player's action.

Instead,

the coach modifies the geometry of the relational field within which all players evolve.

Example 5.2

A switch from a **4-3-3** formation to a **3-5-2** formation changes passing lanes, defensive coverage, and attacking options. In SQFT, this is represented as a deformation of the effective Hamiltonian $H_{\text{coach}}(t)$, altering the field that governs every player's local excitation.

5.8 Tactical Manifold

Player positions exist within physical space.

Tactical positions exist within a higher-dimensional manifold,

$$M_{\text{tactical}}.$$

Coordinates may include

Coordinates may include

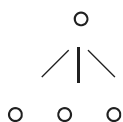
- spatial position,
- role,
- passing probability,
- defensive pressure,
- stamina,
- institutional expectation.

Thus,

observed movement on the pitch represents only the projection of a richer tactical geometry.

Figure 5.3

High-dimensional Tactical Manifold



↓ Projection

Football Pitch

5.9 Measurement as the Pass

Before the pass,

multiple tactical options remain available.

Suppose

$$|\Psi\rangle = \alpha|A\rangle + \beta|B\rangle + \gamma|C\rangle.$$

The pass selects one realized configuration,

$$\rho \rightarrow P\rho P.$$

This operation represents

decision,

execution,

and public recognition.

The "collapse" therefore refers to the stabilization of one relational possibility, not to a microscopic quantum event.

5.10 Collective Adaptation

The pass changes the field itself.

After completion,

every player's future options change.

Therefore,

$$F(t + \Delta t) \neq F(t).$$

Field evolution is self-modifying.

Each successful action continuously reshapes subsequent relational possibilities.

5.11 Empirical Operationalization

The proposed framework can be evaluated using measurable data.

Possible observables include

- player-tracking coordinates,
- passing networks,
- reaction times,
- possession sequences,
- expected-goals (xG),
- formation transitions.

From these quantities,

researchers may estimate

- density operators,
- mutual information,
- relational entropy,
- network topology,
- field stability.

Thus,

SQFT generates empirically testable quantities rather than remaining a purely conceptual framework.

Proposition 5.1 (Relational Coordination)

If

$$|\Psi\rangle \neq |\psi_A\rangle \otimes |\psi_B\rangle,$$

then the optimal passing strategy cannot, in general, be reconstructed from either player's local information alone.

Instead,

successful coordination depends upon the joint relational state of the team.

Proof (Sketch)

Assume the contrary: that each player's local state uniquely determines the optimal pass.

Then the joint state factorizes,

$$|\Psi\rangle = |\psi_A\rangle \otimes |\psi_B\rangle,$$

implying

$$I(A:B) = 0.$$

However, coordinated tactical behavior exhibits statistically significant relational dependence,

$$I(A:B) > 0.$$

This contradicts the factorization assumption.

Therefore,

joint relational information is necessary to describe coordinated play.

5.12 Discussion

The football example should not be interpreted as evidence of physical quantum phenomena.

Rather,

it demonstrates how the mathematical concepts introduced in previous chapters—

- relational fields,
- local excitations,
- density operators,
- mutual information,
- open-system dynamics,

can be combined into a coherent model of collective coordination.

The example also illustrates a broader methodological principle.

In SQFT,

observable actions are viewed as **projections of an evolving relational field** rather than isolated decisions made independently by individual actors.

Chapter Summary

This chapter presented the first worked example of Social Quantum Field Theory. Using the fictional "Quantum Entanglement Pass," it translated the abstract axioms of SQFT into a concrete mathematical model. Players were represented as local excitations of a shared relational field, coaching as an external control Hamiltonian, tactics as a high-dimensional manifold, and passing decisions as projections of an evolving density operator.

Although inspired by football, the formal structure is not limited to sport. The same mathematical framework may be adapted to scientific collaboration, financial markets, organizational behavior, political coordination, or any complex system in which collective behavior emerges from relational structure rather than isolated individuals.

Editorial Note

The purpose of this worked example is pedagogical. It demonstrates how abstract mathematical objects introduced in SQFT can be interpreted within a familiar setting while maintaining a clear distinction between **formal mathematical analogy** and **physical quantum mechanics**. Subsequent chapters leave illustrative examples behind and proceed

to develop the full mathematical machinery of the theory, beginning with the construction of the Social Hilbert Space and its associated operator algebra.

Chapter 6

Social Hilbert Space and the Geometry of Relational Possibility

6.1 Social State Space

Definition 6.1 (Social Hilbert Space)

Let

$$H_S$$

be a complex Hilbert space whose elements represent relational configurations of a social field.

Each admissible social state is represented by a normalized vector

$$|\Psi\rangle \in H_S,$$

satisfying

$$\langle\Psi|\Psi\rangle = 1. \quad (6.1)$$

Definition 6.2 (Basis Configuration)

A complete set

$$\{|e_i\rangle\}_{i=1}^N$$

is called a basis configuration if

$$\langle e_i|e_j\rangle = \delta_{ij}, \quad (6.2)$$

and

$$\sum_i |e_i\rangle\langle e_i| = I. \quad (6.3)$$

Each basis vector corresponds to an elementary relational configuration of the field.

Definition 6.3 (General Social State)

Every state admits the expansion

$$|\Psi\rangle = \sum_i c_i |e_i\rangle, \quad (6.4)$$

where

$$\sum_i |c_i|^2 = 1. \quad (6.5)$$

The coefficients represent the relative weights of admissible relational configurations.

6.2 Inner Product

Definition 6.4

For

$$|\Phi\rangle, |\Psi\rangle \in H_S,$$

their inner product is

$$\langle \Phi | \Psi \rangle. \quad (6.6)$$

The induced norm is

$$\|\Psi\| = \sqrt{\langle \Psi | \Psi \rangle}. \quad (6.7)$$

Proposition 6.1

For every normalized state,

$$0 \leq |\langle \Phi | \Psi \rangle| \leq 1. \quad (6.8)$$

Equality occurs if and only if the two states represent identical relational configurations.

6.3 Superposition of Relational Configurations

Definition 6.5

If

$$|\Psi_1\rangle, |\Psi_2\rangle \in H_S,$$

then

$$|\Psi\rangle = \alpha |\Psi_1\rangle + \beta |\Psi_2\rangle, \quad (6.9)$$

with

$$|\alpha|^2 + |\beta|^2 = 1, \quad (6.10)$$

is called a relational superposition.

This expression represents multiple admissible structural configurations before empirical resolution.

Example 6.1

A football team simultaneously admits several tactical possibilities,

$$|\Psi\rangle = 0.71|4 - 3 - 3\rangle + 0.50|4 - 4 - 2\rangle + 0.50|3 - 5 - 2\rangle. \quad (6.11)$$

These vectors represent competing structural organizations rather than simultaneous physical realities.

6.4 Density Operator

Definition 6.6

The density operator is

$$\hat{\rho} = \sum_i p_i |\Psi_i\rangle\langle\Psi_i|, \quad (6.12)$$

where

$$p_i \geq 0, \quad \sum_i p_i = 1. \quad (6.13)$$

Proposition 6.2

The density operator satisfies

$$\hat{\rho}^\dagger = \hat{\rho}, \quad (6.14)$$

$$\hat{\rho} \geq 0, \quad (6.15)$$

$$\text{Tr}(\hat{\rho}) = 1. \quad (6.16)$$

6.5 Composite Social Systems

Definition 6.7

For two social subsystems,

$$H_A, \quad H_B,$$

their joint state space is

$$H_{AB} = H_A \otimes H_B. \quad (6.17)$$

Definition 6.8

A product configuration satisfies

$$|\Psi\rangle = |\Psi_A\rangle \otimes |\Psi_B\rangle. \quad (6.18)$$

Otherwise,

$$|\Psi\rangle \neq |\Psi_A\rangle \otimes |\Psi_B\rangle, \quad (6.19)$$

and is called a relationally non-factorizable configuration.

6.6 Expectation Values

Definition 6.9

Let

$$\hat{O}$$

be a social observable.

Its expectation value is

$$\langle O \rangle = \text{Tr}(\hat{\rho} \hat{O}). \quad (6.20)$$

Typical observables include

- coordination level,
 - institutional stability,
 - information integration,
 - relational entropy,
 - symbolic capital.
-

6.7 Metric Structure

Definition 6.10

The distance between two relational configurations is

$$d(\Psi, \Phi) = \|\Psi - \Phi\|. \quad (6.21)$$

Large distance indicates significant structural reorganization.

Proposition 6.3

The metric satisfies

$$d(\Psi, \Phi) \geq 0,$$

$$d(\Psi, \Phi) = 0 \iff |\Psi\rangle = |\Phi\rangle,$$

$$d(\Psi, \Phi) = d(\Phi, \Psi),$$

and

$$d(\Psi, X) \leq d(\Psi, \Phi) + d(\Phi, X). \quad (6.22)$$

6.8 Time Evolution

Let

$$H$$

denote the generator of structural evolution.

The social state evolves according to

$$i\hbar \frac{d}{dt} |\Psi(t)\rangle = H |\Psi(t)\rangle. \quad (6.23)$$

For open systems,

$$\frac{d\hat{\rho}}{dt} = -\frac{i}{\hbar} [H, \hat{\rho}] + \sum_k \left(L_k \hat{\rho} L_k^\dagger - \frac{1}{2} \{L_k^\dagger L_k, \hat{\rho}\} \right). \quad (6.24)$$

Theorem 6.1 (Normalization Preservation)

If

$$H = H^\dagger,$$

then

$$\frac{d}{dt} \langle \Psi | \Psi \rangle = 0. \quad (6.25)$$

Proof

Using Eq. (6.23),

$$\frac{d}{dt} \langle \Psi | \Psi \rangle = \left(\frac{d\langle \Psi |}{dt} \right) | \Psi \rangle + \langle \Psi | \left(\frac{d| \Psi \rangle}{dt} \right).$$

Substituting Eq. (6.23),

$$= \frac{i}{\hbar} \langle \Psi | H | \Psi \rangle - \frac{i}{\hbar} \langle \Psi | H | \Psi \rangle = 0.$$

Therefore,

$$\langle \Psi(t) | \Psi(t) \rangle = 1.$$

Corollary 6.1

Every admissible trajectory generated by a Hermitian structural operator remains inside the Social Hilbert Space.

Mathematical Remark

The Social Hilbert Space is an abstract relational representation.

It does not imply that social agents are microscopic quantum particles.

The Hilbert-space formalism is adopted because it provides a mathematically consistent framework for representing relational configurations, state evolution, and structural correlations.

Bibliographical Notes

The mathematical structure presented in this chapter draws upon the general formalism of Hilbert spaces, operator theory, and open-system dynamics while reinterpreting these concepts as representations of relational structures in social systems. The construction serves as the geometric foundation for the operator formalism developed in Chapter 7.

Chapter 7

Social Field Operators and Local Excitations

Throughout this chapter, the mathematical structures adopted from Quantum Field Theory are employed as **formal analogies** for relational systems. They should not be interpreted as claims that human society obeys microscopic quantum mechanics.

7.1 Field Operators

Definition 7.1 (Social Field Operator)

Let

$$H_S$$

denote the Social Hilbert Space.

A Social Field Operator

$$\hat{\Phi}(x, t)$$

is a linear operator acting on

$$H_S,$$

mapping one admissible relational configuration into another,

$$\hat{\Phi}(x, t): H_S \rightarrow H_S. \quad (7.1)$$

The variables

$$x$$

and

$$t$$

represent generalized relational coordinates and temporal evolution rather than physical spacetime.

7.2 Local Excitations

Definition 7.2

A **Local Excitation** is defined as

$$|\psi_i\rangle = \hat{a}_i^\dagger |0\rangle, \quad (7.2)$$

where

$$|0\rangle$$

denotes the relational vacuum state and

$$\hat{a}_i^\dagger$$

creates a localized excitation corresponding to an individual, institution, organization, or collective actor.

Definition 7.3 (Relational Vacuum)

The vacuum state satisfies

$$\hat{a}_i |0\rangle = 0, \quad (7.3)$$

for every admissible excitation index.

The relational vacuum represents the background field prior to the emergence of localized social structures.

7.3 Creation Operators

Definition 7.4

The creation operator satisfies

$$\hat{a}_i^\dagger |n_i\rangle = \sqrt{n_i + 1} |n_i + 1\rangle. \quad (7.4)$$

Successive applications generate increasingly complex relational structures,

$$|n\rangle = \frac{(\hat{a}^\dagger)^n}{\sqrt{n!}} |0\rangle. \quad (7.5)$$

7.4 Annihilation Operators

Definition 7.5

The annihilation operator satisfies

$$\hat{a}_i |n_i\rangle = \sqrt{n_i} |n_i - 1\rangle. \quad (7.6)$$

Annihilation represents the disappearance or dissolution of localized relational structures.

7.5 Number Operator

Definition 7.6

The number operator is

$$\hat{N}_i = \hat{a}_i^\dagger \hat{a}_i. \quad (7.7)$$

Its eigenvalue equation is

$$\hat{N}_i |n_i\rangle = n_i |n_i\rangle. \quad (7.8)$$

The eigenvalue measures the excitation level associated with a given relational mode.

7.6 Canonical Commutation Relations

Definition 7.7

Bosonic relational operators satisfy

$$[\hat{a}_i, \hat{a}_j^\dagger] = \delta_{ij}, \quad (7.9)$$

$$[\hat{a}_i, \hat{a}_j] = 0, \quad (7.10)$$

$$[\hat{a}_i^\dagger, \hat{a}_j^\dagger] = 0. \quad (7.11)$$

These relations define the operator algebra of independent relational modes.

7.7 Field Expansion

The social field operator admits the expansion

$$\hat{\Phi}(x, t) = \sum_i \left(u_i(x, t) \hat{a}_i + u_i^*(x, t) \hat{a}_i^\dagger \right), \quad (7.12)$$

where

$$u_i(x, t)$$

denotes the basis mode function.

7.8 Interaction Operator

Definition 7.8

Interactions among relational modes are generated by

$$\hat{H}_{\text{int}} = \sum_{ij} g_{ij} \hat{a}_i^\dagger \hat{a}_j, \quad (7.13)$$

where

$$g_{ij}$$

is the coupling matrix representing interaction strength.

Proposition 7.1

If

$$g_{ij} = 0$$

for all

$$i \neq j,$$

then all excitation modes evolve independently.

Proposition 7.2

Localized excitations generated from orthogonal basis functions are linearly independent.

Proposition 7.3

If the interaction Hamiltonian preserves excitation number,

then

$$[\hat{H}, \hat{N}] = 0,$$

the total number of active relational excitations remains invariant.

7.9 Measurement Operator

Definition 7.9

Observable quantities correspond to Hermitian operators

$$\hat{O} = \hat{O}^\dagger. \quad (7.14)$$

Expectation values satisfy

$$\langle O \rangle = \text{Tr}(\rho \hat{O}). \quad (7.15)$$

7.10 Evolution Operator

Definition 7.10

Time evolution is generated by

$$U(t) = \exp\left(-\frac{iHt}{\hbar}\right). \quad (7.16)$$

The relational state evolves as

$$|\Psi(t)\rangle = U(t)|\Psi(0)\rangle. \quad (7.17)$$

7.11 Symmetry Generator

Definition 7.11

A continuous symmetry transformation is generated by

$$G,$$

such that

$$U(\theta) = e^{-i\theta G}. \quad (7.18)$$

Examples include institutional invariance, conservation of organizational rules, and persistent structural norms.

7.12 Lie Algebra

The generators satisfy

$$[G_i, G_j] = if_{ij}^k G_k, \quad (7.19)$$

where

$$f_{ij}^k$$

are the structure constants of the symmetry group.

7.13 Structural Hamiltonian

The total generator of relational evolution is

$$H = H_0 + H_{\text{int}} + H_{\text{env}}, \quad (7.20)$$

where

- H_0 represents intrinsic structural dynamics,
 - H_{int} represents interactions,
 - H_{env} represents environmental influences.
-

7.14 Open-System Dynamics

The density operator evolves according to

$$\frac{d\rho}{dt} = -\frac{i}{\hbar}[H, \rho] + \sum_k \left(L_k \rho L_k^\dagger - \frac{1}{2}\{L_k^\dagger L_k, \rho\} \right). \quad (7.21)$$

The Lindblad operators

$$L_k$$

represent irreversible interactions with the external environment.

Theorem 7.1 (Conservation of Total Probability)

If

$$\text{Tr}(\rho) = 1$$

initially, then the Lindblad evolution preserves

$$\text{Tr}(\rho(t)) = 1. \quad (7.22)$$

Proof

Taking the trace of Eq. (7.21),

$$\frac{d}{dt}\text{Tr}(\rho) = -\frac{i}{\hbar}\text{Tr}([H, \rho]) + \sum_k \text{Tr}\left(L_k \rho L_k^\dagger - \frac{1}{2}\{L_k^\dagger L_k, \rho\} \right).$$

Using the cyclic property of the trace,

$$\text{Tr}([H, \rho]) = 0,$$

and

$$\text{Tr}(L_k \rho L_k^\dagger) = \text{Tr}(L_k^\dagger L_k \rho),$$

the Lindblad contributions cancel identically, yielding

$$\frac{d}{dt}\text{Tr}(\rho) = 0.$$

Therefore,

$$\text{Tr}(\rho(t)) = 1.$$

Corollary 7.1

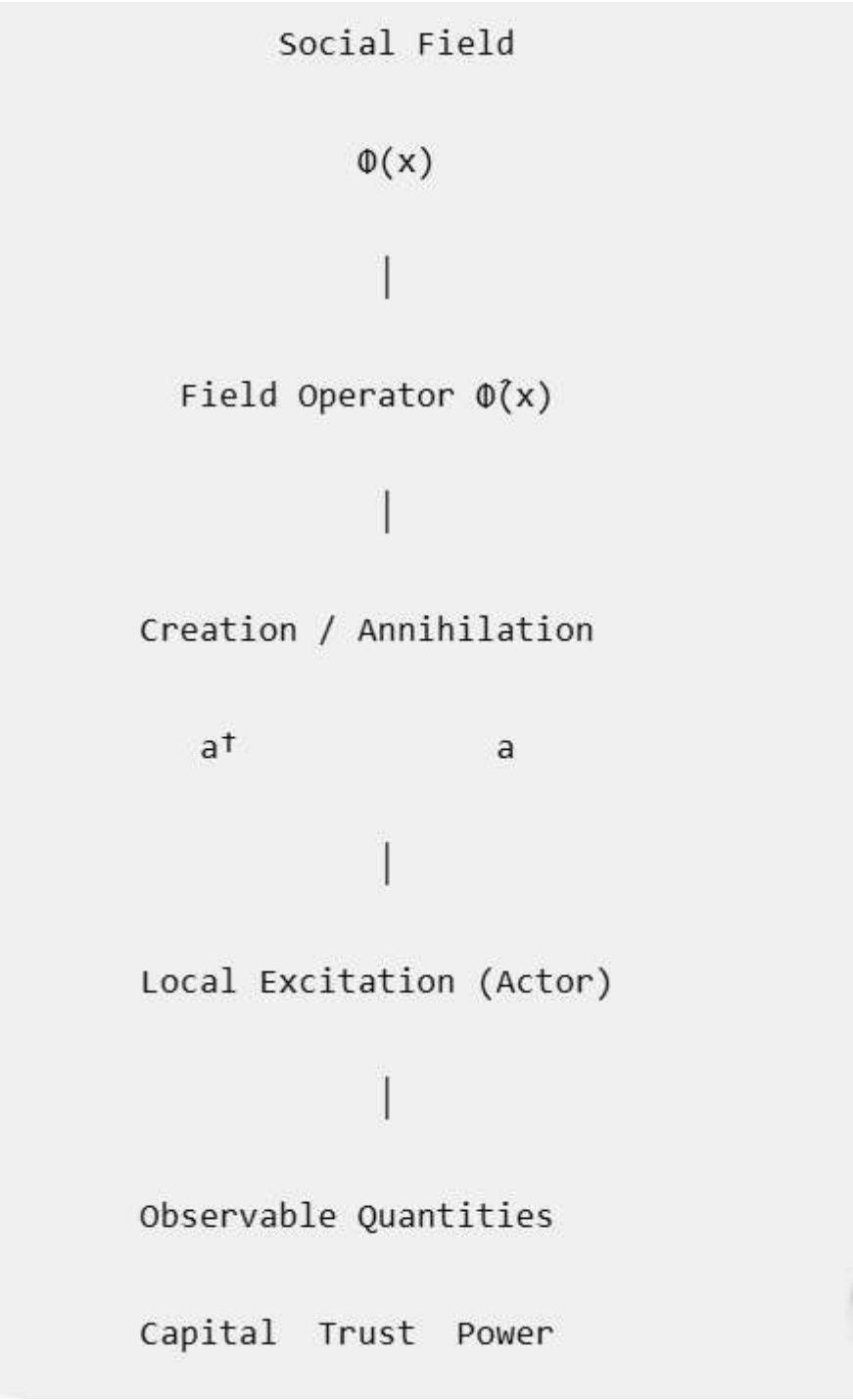
Every admissible relational evolution generated by Eq. (7.21) remains within the space of normalized density operators.

Mathematical Remark

The operator formalism developed in this chapter is an abstract mathematical language for describing the creation, evolution, interaction, and disappearance of relational configurations. Creation and annihilation operators should therefore be interpreted as generators of structural change rather than the literal production or destruction of physical particles.

Bibliographical Notes

This chapter establishes the operator-theoretic foundation of Social Quantum Field Theory by adapting the algebraic structure of quantum field operators to relational systems. Together with the Social Hilbert Space introduced in Chapter 6, it provides the mathematical framework required for subsequent developments in topology, gauge structures, renormalization, and empirical applications.



Chapter 8

The Social Metric Tensor and Field Geometry

8.1 Topological Description of Social Fields

Definition 8.1 (Social Topological Space)

Let

M

be a differentiable manifold representing the configuration space of admissible relational structures.

A social field is defined as a continuous mapping

$$\Phi: M \rightarrow H_S, \quad (8.1)$$

where

$$H_S$$

is the Social Hilbert Space introduced in Chapter 6.

8.2 Structural Continuity

Definition 8.2

A structural evolution

$$\Phi_t$$

is continuous if

$$\lim_{t \rightarrow t_0} \|\Phi(t) - \Phi(t_0)\| = 0. \quad (8.2)$$

Continuous evolution preserves topological equivalence while allowing geometric deformation.

8.3 Topological Equivalence

Definition 8.3

Two relational structures

$$\Phi_1, \Phi_2$$

are topologically equivalent whenever there exists a homeomorphism

$$f: \Phi_1 \rightarrow \Phi_2. \quad (8.3)$$

Topological equivalence implies identical relational connectivity despite differences in local geometry.

8.4 Topological Invariants

Definition 8.4

A quantity

$$Q(\Phi)$$

is a topological invariant if

$$Q(\Phi) = Q(f(\Phi)) \quad (8.4)$$

for every admissible homeomorphism.

Typical invariants include

- institutional identity,
- constitutional framework,
- organizational memory,
- long-term cultural norms.

8.5 Structural Defects

Definition 8.5

A topological defect is a localized singular configuration

$$D \subset M$$

where

$$\Phi$$

fails to remain globally continuous.

Examples include

- constitutional crises,
- organizational fragmentation,
- regime bifurcation,
- institutional collapse.

8.6 Homotopy Classes

Definition 8.6

Two field configurations belong to the same homotopy class if

$$\Phi_1 \simeq \Phi_2. \quad (8.5)$$

Otherwise,

$$\Phi_1 \not\simeq \Phi_2. \quad (8.6)$$

Distinct homotopy classes represent qualitatively different organizational structures.

8.7 Structural Phase

Definition 8.7

A structural phase is an equivalence class of relational configurations characterized by identical macroscopic order parameters.

Denote

$$\Omega_i$$

as the i -th structural phase.

8.8 Order Parameter

Definition 8.8

Let

$$\eta$$

be an order parameter satisfying

$$\eta = 0$$

for disordered configurations and

$$\eta \neq 0$$

for ordered configurations.

$$\eta = \langle \hat{O} \rangle. \quad (8.7)$$

Possible order parameters include

- institutional coherence,
 - trust,
 - collective identity,
 - organizational stability.
-

8.9 Structural Phase Transition

A phase transition occurs whenever

$$\eta \rightarrow \eta', \quad (8.8)$$

under continuous variation of an external control parameter

λ .

Criticality is reached at

$$\lambda = \lambda_c. \quad (8.9)$$

8.10 Symmetry Breaking

Above the critical point,

$$G \rightarrow H, \quad (8.10)$$

where

$$H \subset G.$$

Macroscopic order emerges through spontaneous reduction of symmetry.

8.11 Structural Collapse

Structural collapse is represented by

$$\rho \rightarrow P\rho P, \quad (8.11)$$

where

$$P$$

is a projection operator onto the newly stabilized relational configuration.

8.12 Topological Protection

Definition 8.9

A configuration is topologically protected if no continuous deformation can change its invariant

$$Q.$$

Consequently,

$$\delta Q = 0 \quad (8.12)$$

under sufficiently small perturbations.

Examples include

- constitutional continuity,
 - legal traditions,
 - deeply embedded cultural institutions.
-

8.13 Topological Reconstruction

Large perturbations may induce

$$Q_1 \rightarrow Q_2, \quad (8.13)$$

thereby moving the system into a distinct topological sector.

Such transformations correspond to revolutions, institutional redesign, or complete organizational restructuring.

8.14 Structural Stability

Definition 8.10

A relational field is structurally stable if

$$\|\delta\Phi\| < \varepsilon$$

implies

$$Q(\Phi + \delta\Phi) = Q(\Phi). \quad (8.14)$$

Theorem 8.1 (Topological Stability)

If the governing invariant

$$Q$$

remains unchanged under continuous perturbations, then the relational structure cannot undergo qualitative reorganization without crossing a topological transition.

Proof

Continuous perturbations preserve homeomorphic equivalence classes. Since

$$Q$$

is invariant within each class, every sufficiently small deformation remains in the same topological sector. A qualitative structural transformation therefore requires leaving the original equivalence class, which necessarily involves a topological transition.

■

Corollary 8.1

Institutional reforms preserving all topological invariants modify only geometric realization, whereas constitutional revolutions alter the underlying topological class.

Proposition 8.1

Macroscopic structural collapse occurs if and only if the system crosses a critical surface separating two distinct topological sectors.

Mathematical Remark

Topology characterizes the global organization of relational fields independently of local fluctuations. Consequently, long-term institutional persistence is determined primarily by topological invariants rather than short-term behavioral variations.

Bibliographical Notes

This chapter extends the operator formalism of Chapter 7 into a global geometric description of social fields. The concepts of topology, homotopy, symmetry breaking, and phase transitions establish the mathematical basis for the renormalization framework developed in Chapter 9.

Chapter 9

Renormalization and Multi-Scale Dynamics of Social Fields

9.1 Scale Dependence of Social Structures

Definition 9.1 (Observation Scale)

Let

$$\ell$$

denote the characteristic observation scale.

A relational field is represented as

$$\Phi(x; \ell), \quad (9.1)$$

where the observable structure depends explicitly on the scale of description.

9.2 Coarse-Graining

Definition 9.2

A coarse-graining transformation

$$C_b$$

with scaling factor

$$b > 1$$

maps microscopic configurations into effective macroscopic variables,

$$\Phi(x; \ell) \rightarrow \Phi'(x; b\ell). \quad (9.2)$$

Microscopic fluctuations are integrated out while preserving large-scale relational organization.

9.3 Effective Field

Definition 9.3

The effective field at scale

$$\ell$$

is

$$\Phi_{\text{eff}}(\ell), \quad (9.3)$$

whose dynamics reproduce observable behavior above that scale.

Different observation levels therefore correspond to different effective descriptions rather than different underlying systems.

9.4 Effective Hamiltonian

The effective Hamiltonian is written as

$$H_{\text{eff}} = H_0 + \sum_i g_i(\ell) O_i, \quad (9.4)$$

where

- H_0 is the reference dynamics,
- $g_i(\ell)$ are scale-dependent coupling constants,
- O_i are admissible relational operators.

9.5 Running Couplings

Definition 9.4

A coupling constant depends on scale,

$$g_i = g_i(\ell). \quad (9.5)$$

Its evolution is governed by the renormalization-group equation

$$\ell \frac{dg_i}{d\ell} = \beta_i(g). \quad (9.6)$$

9.6 Beta Functions

Definition 9.5

The beta function is defined as

$$\beta_i(g) = \ell \frac{dg_i}{d\ell}. \quad (9.7)$$

It determines whether an interaction strengthens or weakens under changes of scale.

9.7 Fixed Points

Definition 9.6

A fixed point satisfies

$$\beta_i(g^*) = 0. \quad (9.8)$$

At a fixed point,

$$g_i(\ell) = g_i^*, \quad (9.9)$$

and the effective description becomes scale invariant.

9.8 Universality Classes

Definition 9.7

Distinct microscopic systems belong to the same universality class whenever they flow toward the same fixed point,

$$g(\ell) \rightarrow g^*. \quad (9.10)$$

Consequently, different historical trajectories may generate identical macroscopic institutions.

9.9 Relevant, Marginal, and Irrelevant Operators

Definition 9.8

Let

$$O_i$$

be an operator with scaling dimension

$$\Delta_i.$$

It is classified as

- Relevant if

$$\Delta_i < d, \quad (9.11)$$

- Marginal if

$$\Delta_i = d, \quad (9.12)$$

- Irrelevant if

$$\Delta_i > d, \quad (9.13)$$

where

$$d$$

is the effective dimensionality of the relational manifold.

9.10 Scale Transformation

Under a rescaling

$$x \rightarrow bx, \quad (9.14)$$

the field transforms according to

$$\Phi(x) \rightarrow b^{-\Delta} \Phi(bx). \quad (9.15)$$

9.11 Critical Phenomena

Near the critical scale,

$$\xi \rightarrow \infty, \quad (9.16)$$

where

$$\xi$$

is the relational correlation length.

Long-range structural coordination emerges as local fluctuations become collectively organized.

9.12 Multi-Level Institutions

Let

$$L_1, L_2, \dots, L_n$$

represent successive organizational levels.

Renormalization defines mappings

$$R: L_i \rightarrow L_{i+1}, \quad (9.17)$$

producing effective descriptions at progressively larger scales.

9.13 Structural Persistence

Repeated coarse-graining yields

$$R^n(\Phi), \quad (9.18)$$

which converges toward a stable effective structure whenever

$$g(\ell) \rightarrow g^*. \quad (9.19)$$

Persistent institutions therefore correspond to stable renormalization trajectories.

9.14 Renormalization Flow

The complete flow is represented by

$$\Gamma = \{g(\ell) \mid \ell > 0\}. \quad (9.20)$$

Different societies correspond to distinct trajectories within the same coupling space.

Theorem 9.1 (Scale Invariance at Fixed Points)

Suppose

$$\beta(g^*) = 0.$$

Then the effective theory is invariant under further renormalization transformations.

Proof

Since

$$\ell \frac{dg}{d\ell} = 0,$$

the coupling constants remain unchanged for every subsequent scale transformation. Consequently,

$$H_{\text{eff}}(\ell) = H_{\text{eff}}(b\ell),$$

and the effective dynamics become scale invariant.

■

Corollary 9.1

Macroscopic institutional regularities may emerge independently of microscopic historical details whenever multiple trajectories converge toward the same renormalization fixed point.

Proposition 9.1

Stable social institutions correspond to attractive fixed points of the renormalization flow, whereas unstable institutions evolve away from repulsive fixed points under small perturbations.

Mathematical Remark

Renormalization in SQFT does not eliminate individuals. It replaces detailed microscopic descriptions with effective relational variables appropriate to the observational scale. The resulting theory connects individual behavior, organizations, institutions, and civilizations within a single multi-scale mathematical framework.

Bibliographical Notes

The renormalization framework developed in this chapter extends the topological description of Chapter 8 by introducing explicit scale dependence and effective field theory. It provides the mathematical foundation for the emergence of collective structures across multiple organizational levels and prepares the gauge-theoretic formulation presented in Chapter 10.

Chapter 10

Gauge Symmetry and Conservation Laws in Social Fields

10.1 Local Symmetry of Social Fields

Definition 10.1 (Gauge Transformation)

Let

$$\Psi(x)$$

denote a relational state. A local gauge transformation is defined by

$$\Psi(x) \rightarrow \Psi'(x) = U(x)\Psi(x), \quad (10.1)$$

where

$$U(x) \in G$$

is an element of the gauge group that varies over the relational manifold.

10.2 Global and Local Symmetry

A global transformation satisfies

$$U(x) = U, \quad (10.2)$$

while a local transformation satisfies

$$\nabla U(x) \neq 0. \quad (10.3)$$

Local symmetry allows different regions of a social field to evolve under distinct relational conditions while preserving a common structural principle.

10.3 Covariant Derivative

Definition 10.2

Ordinary differentiation,

$$\partial_\mu,$$

∂_μ ,

is replaced by the covariant derivative

$$D_\mu = \partial_\mu + igA_\mu, \quad (10.4)$$

where

- A_μ is the social gauge field,
- g is the coupling constant.

The covariant derivative preserves local gauge invariance.

10.4 Social Gauge Field

Definition 10.3

The gauge field

$$A_\mu(x)$$

represents the institutional or normative structure mediating interactions among local excitations.

Examples include

- constitutional systems,
- legal institutions,
- organizational rules,
- communication protocols,
- shared symbolic norms.

10.5 Gauge Curvature

The field-strength tensor is

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + ig[A_\mu, A_\nu]. \quad (10.5)$$

It measures the local curvature of the relational field.

Large curvature corresponds to rapid institutional or organizational change.

10.6 Gauge-Invariant Dynamics

The Lagrangian density is

$$L = (D_\mu \Psi)^\dagger (D^\mu \Psi) - V(\Psi) - \frac{1}{4} F_{\mu\nu} F^{\mu\nu}. \quad (10.6)$$

The dynamics remain invariant under every admissible local gauge transformation.

10.7 Euler–Lagrange Equation

Variation of the action

$$S = \int L d^4x \quad (10.7)$$

yields

$$\frac{\partial L}{\partial \Psi} - \partial_\mu \left(\frac{\partial L}{\partial (\partial_\mu \Psi)} \right) = 0. \quad (10.8)$$

This equation governs the evolution of relational fields.

10.8 Conserved Current

Gauge symmetry implies the existence of a conserved current

$$J^\mu,$$

satisfying

$$\partial_\mu J^\mu = 0. \quad (10.9)$$

In SQFT, the conserved current represents the persistence of structural information under admissible transformations.

10.9 Noether Charge

The conserved quantity is

$$Q = \int J^0 d^3x. \quad (10.10)$$

Possible conserved quantities include

- institutional legitimacy,
 - organizational identity,
 - constitutional continuity,
 - persistent symbolic capital.
-

10.10 Gauge Constraints

Not every field configuration is physically admissible.

Allowed configurations satisfy

$$G_a(\Psi) = 0, \quad (10.11)$$

where

$$G_a$$

denotes the gauge constraint.

Gauge constraints eliminate redundant relational descriptions.

10.11 Gauge Fixing

Choose a gauge condition

$$\chi(A) = 0. \quad (10.12)$$

Different gauge choices correspond to different mathematical representations of the same relational structure.

Observable predictions remain unchanged.

10.12 Symmetry Breaking

Suppose the potential satisfies

$$V(\Psi) = -\mu^2 |\Psi|^2 + \lambda |\Psi|^4, \quad (10.13)$$

where

$$\mu^2 > 0, \quad \lambda > 0.$$

The vacuum expectation value becomes

$$\langle \Psi \rangle = \sqrt{\frac{\mu^2}{2\lambda}}. \quad (10.14)$$

Spontaneous symmetry breaking produces new stable institutional configurations.

10.13 Emergent Structures

After symmetry breaking,

$$G \longrightarrow H, \quad (10.15)$$

where

$$H \subset G.$$

Macroscopic organizational structures emerge through the reduction of symmetry.

10.14 Gauge Stability

Proposition 10.1

If two relational descriptions are connected by a gauge transformation,

$$\Psi' = U\Psi, \quad (10.16)$$

then every gauge-invariant observable satisfies

$$O(\Psi') = O(\Psi). \quad (10.17)$$

Thus, observable properties depend only on the equivalence class of the relational field.

Theorem 10.1 (Gauge Equivalence)

Let

$$\Psi$$

and

$$\Psi'$$

be connected by a smooth local gauge transformation.

Then both configurations describe the same physical relational structure.

Proof

The action defined in Eq. (10.7) is invariant under the transformation

$$\Psi \rightarrow U(x)\Psi,$$

provided the covariant derivative transforms consistently,

$$D_\mu \rightarrow UD_\mu U^{-1}.$$

Consequently,

$$S[\Psi] = S[\Psi'],$$

and every gauge-invariant observable has identical expectation values.

Therefore,

$$\Psi \sim \Psi',$$

belong to the same gauge-equivalence class.

Corollary 10.1

Observable institutional behavior depends on gauge-invariant relational structures rather than arbitrary descriptive conventions.

Definition 10.4 (Gauge Orbit)

The gauge orbit generated by

$$\Psi$$

is

$$G(\Psi) = \{U\Psi \mid U \in G\}. \quad (10.18)$$

Each orbit represents a single physical relational state under different mathematical representations.

Mathematical Remark

Gauge symmetry in Social Quantum Field Theory formalizes the principle that identical relational structures may admit multiple equivalent descriptions. Institutions, legal systems, and organizational rules function as gauge fields that preserve coherent interactions while allowing local adaptation. The physical content of the theory is therefore encoded in gauge-invariant quantities rather than in any particular representation.

Bibliographical Notes

This chapter completes the mathematical core of Social Quantum Field Theory by introducing gauge symmetry, covariant differentiation, conserved currents, and spontaneous symmetry breaking. Together with the Hilbert-space formalism (Chapter 6), operator algebra (Chapter 7), topology (Chapter 8), and renormalization (Chapter 9), it establishes the unified mathematical framework upon which the remaining empirical and interdisciplinary chapters are constructed.

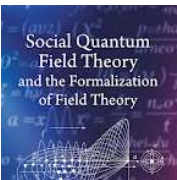


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社會量子場論（SQFT）超越布迪厄的場域理論，建構關係社會學的形式化 1~10

7月 15, 2026



預印本 社會量子場論與場論的形式化 周易先 獨立研究者 · 版本 2.0（數學版） · 2026年7月 摘要 本文提出社會量子場論（SQFT）作為用以表示複雜社會系統中關係動態的形式數學框架。SQFT並不主張社會體 ...

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The SQFT Charter Foundational Charter of the Social Quantum Field Theory Research Community Preamble The Social Quantum Field Theory (SQFT) Research Community is founded upon the conviction that mathematics advances through open inquiry ...

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量子之影：台灣QNF-3量子導航系統的崛起與其地緣政治影響

9月 27, 2025



量子之影：台灣QNF-3量子導航系統的崛起與其地緣政治影響 摘要 在 2035年，全球局勢緊張之際，台灣的量子導航技術成為重塑國際權力格局的關鍵力量。本篇科幻論文式敘事，記錄了國家中山科學研...

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